

1. Properties of Variance

Variance is the average *of the square* of the distance from the mean. For this reason, variance is sometimes called the “mean square deviation.” Then we take its square root to get the standard deviation—which in turn is called “root mean square deviation.”

But why bother squaring? Why not study the actual distance from the mean, namely, the absolute value of $R - \mathbb{E}[R]$, instead of its root mean square? The answer is that variance and standard deviation have useful properties that make them much more important in probability theory than average absolute deviation.

1.1. A Formula for Variance

Applying linearity of expectation to the formula for variance yields a convenient alternative formula

Lemma 19.3.1.

$$\text{Var}[R] = \mathbb{E}[R^2] - (\mathbb{E}[R])^2 \quad (1)$$

for any random variable, R .

A simple and very useful formula for the variance of an indicator is an immediate consequence.

Corollary 19.3.2. *If B is a Bernoulli variable where $p := \Pr[B = 1]$, then*

$$\text{Var}[B] = p - p^2 = p(1 - p) \quad (2)$$

1.2. Variance of Time to Failure

According to §18.4.6, the mean time to failure is $\frac{1}{p}$ for a process that fails during any given hour with probability p . What about the variance?

By Lemma 19.3.1,

$$\text{Var}[C] = \mathbb{E}[C^2] - \left(\frac{1}{p}\right)^2 \quad (3)$$

so all we need is a formula for $\mathbb{E}[C^2]$.

1.3. Dealing with Constants

It helps to know how to calculate the variance of $aR + b$:

Theorem 19.3.4. [Square Multiple Rule for Variance] *Let R be a random variable and a a constant. Then*

$$\text{Var}[aR] = a^2 \text{Var}[R] \quad (4)$$

It's even simpler to prove that adding a constant does not change the variance.

Theorem 19.3.5. *Let R be a random variable and b a constant. Then*

$$\text{Var}[R + b] = \text{Var}[R] \quad (5)$$

Recalling that the standard deviation is the square root of variance, this implies that the standard deviation of $aR + b$ is simply $|a|$ times the standard deviation of R .

Corollary 19.3.6.

$$\sigma_{aR+b} = |a|\sigma_R \quad (6)$$

1.4. Variance of a Sum

In general, the variance of a sum is not equal to the sum of the variances, but variances do add for *independent* variables. In fact, *mutual* independence is not necessary: *pairwise* independence will do. This is useful to know because there are some important situations, such as Birthday Matching in §16.4, that involve variables that are pairwise independent but not mutually independent.

Theorem 19.3.7. *If R and S are independent random variables, then*

$$\text{Var}[R + S] = \text{Var}[R] + \text{Var}[S] \quad (7)$$

Theorem 19.3.8. *[Pairwise Independent Additivity of Variance] If $R_1, R_2, \dots, R_n$ are pairwise independent random variables, then*

$$\text{Var}[R_1 + R_2 + \dots + R_n] = \text{Var}[R_1] + \text{Var}[R_2] + \dots + \text{Var}[R_n] \quad (8)$$

Now we have a simple way of computing the variance of a variable, J , that has an (n, p) -binomial distribution. We know that $J = \sum_{k=1}^n I_k$ where I_k are the mutually independent indicator variables with $\Pr[I_k = 1] = p$. The variance of each I_k is $p(1 - p)$ by corollary 19.3.2, so by linearity of variance, we have

Lemma 19.3.9 (Variance of the Binomial Distribution). *If J has the (n, p) binomial distribution, then*

$$\text{Var}[J] = n \text{Var}[I_k] = np(1 - p) \quad (9)$$